CHEMISTRY

1) Two elements A & B form compounds having formula AB_2 & AB_4. When dissolved in 20 g of benzene, 1 g of AB_2 lowers the freezing point by 2.3 K whereas 1 g of AB_4 lowers it by 1.3 K. Calculate atomic masses of A & B ? ($K_{f(benzene)}$ = 5.1 K kg $mole^{-1}$)

(1) A = 25.58 ; B = 42.64
(2) A = 42.64 ; B = 25.58
(3) A = 12.79 ; B = 21.32
(4) A = 21.32 ; B = 12.79

2) Vapour pressure over the solution of two volatile liquid is :-

(1) Linearly varies with the mole fraction
(2) Inversely varies with the mole fraction
(3) Exponentially varies with the mole fraction
(4) Independent of mole fraction

3) 100 g of pure liquid A (Mw = 140 g mol^{-1}) was dissolved in 1000 g of pure liquid B (Mw = 180 g mol^{-1}) The vapour pressure of pure B was found to be 500 torr calculate vapour pressure of pure liquid A If the total V.P of solution is 475 torr?

(1) 100 torr
(2) 140 torr
(3) 280 torr
(4) 200 torr

4) Dissolution of gas in water is:-

(1) Exothermic process
(2) Endothermic process
(3) $\Delta H = 0$
(4) $\Delta G = 0$

5) At equilibrium, vapour phase will be rich in the component which is (take, $x_A = x_B$) :-

(1) Less volatile
(2) More volatile
(3) Whose P° is less
(4) None

6) Which one of the following method is used to determine molar mass of proteins, polymers & other

macromolecules?

(1) Relative lowering in Vapour Pressure
(2) Osmotic pressure
(3) Elevation in Boiling Point
(4) Depression in Freezing Point

7) In $\Delta T_f = K_f m$, K_f is called :-

(1) Molal elevation constant
(2) Cryoscopic constant
(3) Van't hoff factor
(4) None of these

8) K_b equals to :-
[T_b, M and ΔH_{vap} are boiling point, molar mass and enthalpy of vaporization for solvent respectively]

(1) $\frac{R \Delta H_{vap} M}{1000 T_b^2}$
(2) $\frac{R T_b^2 M}{1000 \Delta H_{vap}}$
(3) $\frac{\Delta H_{vap} M}{1000 R T_b^2}$
(4) $\frac{R T_b^2 M}{100 \Delta H_{vap.}}$

9) Which of the following are correct ?

(1) $T_b - T_b^o = K_b m$
(2) $T_f^o - T_f = K_f m$
(3) $M_2 = \frac{w_2 RT}{\pi V}$
(4) All of these

10) Reverse osmosis is :-

(1) Flow of solute particles from low conc. to high conc.
(2) Flow of solvent particles from high conc. to low conc.
(3) Flow of solute particles from high conc. to low conc.
(4) Flow of solvent particles from low conc. to high conc.

11) Vapour pressure (V.P.) of solution containing non-volatile solute is always :-

(1) Greater than V.P of pure solvent
(2) Less than V.P of pure solvent

(3) May be greater or may be less than V.P. of pure solvent

(4) Can't be compared without given numerical values.

12) The depression in freezing point of water observed for the same amount of acetic acid, trichloroacetic acid and trifluoro acetic acid increases in the order

(1) Trichloroacetic acid < Acetic acid < Trifluoroacetic acid

(2) Acetic acid < Trichloroacetic acid < Trifluoroacetic acid

(3) Acetic acid < Trifluoroacetic acid < Trichloroacetic acid

(4) None

13)

The vapour pressure of water is 12.3 KPa at 300 K. Vapour pressure of one molal solution of a non-volatile solute in it is:-

(1) 8 KPa

(2) 12.08 KPa

(3) 15 KPa

(4) 18 KPa

14) Azeotropes arise due to :-

(1) Low deviations from Raoult's law

(2) Very large deviations from Raoult's law

(3) Ideal solution

(4) Very low deviations from Raoult's law

15) Which of the following expression(s) is/are correct ?

(1) $i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$

(2) $i = \frac{\text{Total number of particles after dissociation}}{\text{Total number of particles before dissociation}}$

(3) $i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$

(4) All of the above

16) 0.6 mL of CH_3COOH (density = 1.06 g/mL) is dissolved in 1 litre of water. The depression in Freezing point observed for the strength of this acid was 0.0205°C. Calculate the van't hoff factor & Dissociation constant of acid respectively?

(1) 0.0197, 1.86×10^{-5}

(2) 1.041, 1.86×10^{-5}

(3) 1.86×10^{-5}, 1.041

(4) 0.0197, 186×10^{-5}

17) Saturated solution is one in which :-

(1) More solute can be dissolved
(2) No more solute can be dissolved
(3) Rate of dissolution = rate of crystallisation
(4) Both (2) & (3)

18) What %(v/v) solution of ethylene glycol is used for cooling the engine in cars?

(1) 53%
(2) 45%
(3) 35%
(4) 55%

19) Ethanoic acid dimerise in benzene. It normally happens in solvents with :-

(1) High dielectric constant
(2) Low dielectric constant
(3) Both (1) and (2)
(4) None

20) Amalgam of mercury with sodium is an example of :-

(1) Liquid in Solid
(2) Gas in Solid
(3) Solid in Solid
(4) Solid in liquid

21) Which of the following set does not dissolve in water ?

(1) Naphthalene & sugar
(2) NaCl & sugar
(3) Naphthalene & Anthracene
(4) Anthracene & NaCl

22) Calculate the mass of ascorbic acid ($C_6H_8O_6$) to be dissolved in 75 g of acetic acid to lower its freezing point by 1.5°C.
(Given K_f = 3.9 K kg/mol)

(1) 50.76 g
(2) 5.076 g
(3) 0.028 g
(4) 0.28 g

23) Boiling point of solution containing non-volatile solute is always :-

(1) Less than boiling point of pure solvent
(2) Greater than boiling point of pure solvent
(3) Equal to boiling point of pure solvent
(4) Can't be compare to pure solvent

24) 19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in freezing point of water is 1°C. Calculate the van't Hoff factor for CH_2FCOOH ?

(1) 0.075
(2) 1.075
(3) 1.75
(4) 1.57

25) The tanks used by scuba divers contain :-

(1) 11.7% Helium
(2) 56.2% Nitrogen
(3) 32.1% Oxygen
(4) All of these

26) Cellulose acetate sheet is :-

(1) Permeable to ions present in sea water
(2) Permeable to water
(3) Impermeable to water
(4) None

27) In Azeotropic mixture of nitric acid & water (% by mass) :-

(1) Nitric acid – 95%, water – 5%
(2) Nitric acid – 5%, water – 95%
(3) Nitric acid – 68%, water – 32%
(4) Nitric acid – 32%, water – 68%

28) "Aquatic species are more confortable in cold water rather than warm water". It is due to :-

(1) Solubility of gas increases with decrease in temperature
(2) Solubility of gas decreases with decrease in temperature
(3) Solubility of gas increases with increases in temperature
(4) None of these

29) Decrease in V.P of 1kg water is same in :-

(1) 1 mole of urea
(2) 1 mole of Glucose
(3) 1 mole of NaCl

(4) Both (1) & (2)

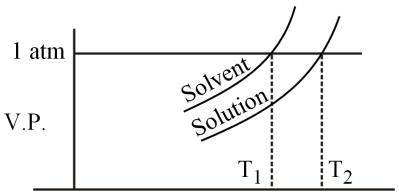
30) 18 g of Glucose is dissolved in $\frac{1}{2}$ kg of water. At what temperature will water boil at 1.013 bar? $(K_b)_{water}$ = 0.52 K kg/mol

(1) 373.0104 K

(2) 373.402 K

(3) 373.254 K

(4) 373.020 K



31) In the fig, T_1 & T_2 are respectively :- [B.P. = Boiling point, F.P. Freezing point]

(1) T_1 = B.P. of Solvent
T_2 = B.P. of Solution

(2) T_1 = B.P. of Solution
T_2 = B.P. of Solvent

(3) T_1 = F.P. of Solvent
T_2 = F.P. of Solution

(4) T_1 = F.P. of Solution
T_2 = F.P. of Solvent

32) Freezing point (F.P.) of solution containing non-volatile solute is always :-

(1) Greater than F.P. of pure solvent

(2) Less than F.P of pure solvent

(3) Equal to F.P of pure solvent

(4) May be greater or less than pure solvent

33) In $\Delta T_b = K_b m$, K_b is called :-

(1) Boiling point elevation constant

(2) Ebullioscopic constant

(3) Both (1) & (2)

(4) Cryoscopic constant

34) Camphor in Nitrogen gas is an example of :-

(1) Liquid in Gas

(2) Gas in Gas

(3) Gas in Liquid

(4) Solid in Gas

35) K_f equal to :-

(1) $\frac{RT_f^2M}{1000\Delta H_{fus}}$

(2) $\frac{R\Delta H_{fus}M}{1000T_f^2}$

(3) $\frac{\Delta H_{fus}M}{1000RT_f^2}$

(4) None of these

36) Which of the following is/are correct for dilute solution :-
(1 → solvent, 2 → solute)
[P_s = Vapour pressure of solution]

(1) $\frac{P_1^0 - P_s}{P_1^0} = \frac{w_2M_1}{w_1M_2}$

(2) $\frac{P_1^0 - P_s}{P_1^0} = \frac{n_1}{n_2}$

(3) $\frac{P_1^0 - P_s}{P_1^0} = \frac{n_1 - n_2}{n_1}$

(4) $\frac{P_s - P_1^o}{P_s} = \frac{n_2}{n_1}$

37) Copper dissolved in gold is an example of :-

(1) Gas in Solid

(2) Solid in liquid

(3) Solid in Solid

(4) Liquid in Solid

38)

Osmosis is :-

(1) Flow of solute particles from low conc. to high conc.

(2) Flow of solvent particles from high conc. to low conc.

(3) Flow of solute particles from high conc. to low conc.

(4) Flow of solvent particles from low conc. to high conc.

39) An aqueous solution of 2% non-volatile solute exerts a pressure of 1.004 bar at the normal boiling point of solvent? Calculate the molar mass of the solute ?

(1) 60.24 g mol^{-1}

(2) 20.24 g mol^{-1}

(3) 41.34 g mol^{-1}

(4) 30.30 g mol^{-1}

40) Unit of K_f is :-

(1) $K\frac{kg}{mol}$

(2) Mole Kg^{-1} K^{-1}

(3) Mole Kg K^{-1}

(4) Kg $Mole^{-1}$ K^{-1}

41) Value of 'i' for NaCl is 1.87 at 0.1 m & 1.94 at 0.01 m. This is due to _____ ?

(1) Association of NaCl

(2) Complete dissociation of NaCl

(3) Incomplete dissociation of NaCl

(4) None of these

42) In $\pi = CRT$, R equals to :-

(1) $0.0821\frac{J}{mol\,K}$

(2) $8.314\frac{cal}{mol\,K}$

(3) $0.0821\frac{lit\,atm}{mol\,K}$

(4) $8.314\frac{lit\,atm}{mol\,K}$

43) Solubility of substance depends on :-

(1) Nature of solute

(2) Nature of solvent

(3) Temperature

(4) All of these

44) In Azeotropic mixture of ethanol-water mixture :-

(1) Ethanol – 95%, water – 5% by volume

(2) Ethanol – 5%, water – 95% by volume

(3) Ethanol – 68%, water – 32% by volume

(4) Ethanol – 32%, water – 68% by volume

45) What % by mass of sodium hypochlorite in water is used in commercial bleaching solution?

(1) 6.32

(2) 3.62
(3) 2.63
(4) None

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	3	1	2	2	2	2	4	2	2	2	2	2	4	2	4	3	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	2	2	4	2	3	1	4	3	1	2	3	4	1	1	3	4	3	1
Q.	41	42	43	44	45															
A.	3	3	4	1	2															

SOLUTIONS

CHEMISTRY